Dynamical Systems and Agent-Based Modelling: Numerical Investigation of the Duffing Oscillator and a Spatial Prey–Predator Model

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I. Introduction

This report addresses three assignment questions. Question 1 (Section II) treats the forced Duffing oscillator, a classical testbed for nonlinear dynamics and chaos [1]: we reduce it to a first-order system, simulate it at $\gamma = 0.29$, justify the discretisation step, and then show that at $\gamma = 0.5$ the system is chaotic via three independent signatures. Question 2 (Section III) constructs an agent-based prey–predator model on a 20×20 grid, an approach widely used in ecological modelling [2], contrasts it with the deterministic Lotka–Volterra ODE [3], and analyses how predator extinction probability depends on the background death rate α. Question 3 (Section IV) extends the Lotka–Volterra system with a predator harvesting term, contrasts constant-yield and constant-effort formulations, and reviews the literature on prey–predator model variants.

II. Question 1: The Duffing Oscillator

A. First-Order Reduction

To convert the Duffing oscillator into a system of first-order differential equations suitable for numerical simulation, we introduced state variables $x_1 := x$ and $x_2 := dx/dt$. The original second-order equation

$$\frac{d^2x}{dt^2} = \gamma\cos(\omega t) - \delta\frac{dx}{dt} - \alpha x - \beta x^3, \tag{1}$$

with $\delta = 0.3$, $\alpha = -1$, $\beta = 1$, $\omega = 1.2$, and γ as the bifurcation parameter, with the substituted state variables becomes:

$$\dot{x}_1 = x_2, \tag{2}$$

$$\dot{x}_2 = \gamma\cos(\omega t) - \delta x_2 - \alpha x_1 - \beta x_1^3, \tag{3}$$

This is the form $\dot{\mathbf{y}} = f(t, \mathbf{y})$ required by standard numerical integrators [4]. Setting $\gamma = 0$ and $\delta = 0$ in eq. (1) and using $\ddot{x} = -dV/dx$ gives

$$\frac{dV}{dx} = \alpha x + \beta x^3, \tag{4}$$

which integrates to

$$V(x) = -\frac{1}{2}x^2 + \frac{1}{4}x^4. \tag{5}$$

Setting $V'(x) = -x + x^3 = x(x^2 - 1) = 0$ gives equilibria at $x^* \in \{-1, 0, +1\}$. Since $V''(\pm 1) = +2 > 0$ and $V''(0) = -1 < 0$, the system has two stable wells at $x^* = \pm 1$ and an unstable saddle at the origin, which is why Figure 3 shows trajectories settling around $x^* = \pm 1$.

B. Autonomous vs. Non-Autonomous

The eqs. (2) to (3) show that the system is *non-autonomous*. This is because the right-hand side of eq. (3) explicitly depends on time t through the forcing term $\gamma\cos(\omega t)$. This specific term alone is what causes a break in autonomy. However, if we set $\gamma = 0$, the forcing term becomes $0 \cdot \cos(\omega t) = 0$, which removes the time dependence entirely. This would then make the system autonomous and only depend on the state variables (x_1, x_2) [1]. Furthermore, we decided to implement the forcing phase $\theta := \omega t \bmod 2\pi$ as a third state variable. This means that eqs. (2) to (3) becomes a 3D autonomous system, which can then be used to construct the Poincaré sections shown in Section II-E.

C. Discretisation Step Selection

To select the correct step size for accurate simulation, we first decided on a reference solution from which to compare all other RK4 solutions. We set the reference at $h_{\text{ref}} = 10^{-4}$ and computed its RK4 solution values. Following this, we then computed RK4 solutions at five different step sizes $h \in \{0.1, 0.05, 0.01, 0.005, 0.002\}$. Finally, we measured the root-mean-squared error (RMSE) between each respective solution and the reference solution. RK4 has a theoretical global error of $\mathcal{O}(h^4)$ [4], so plotting RMSE against h on log–log axes should give a slope of approximately 4. The results are shown in Figure 1.

As shown above in the top panel of Figure 1, five clearly separated error bands emerge. As expected, the highest pointwise error was reached by $h = 0.1$ at ~ 0.8 by $t = 400$. On the other hand, $h = 0.005$ stays below $\sim 10^{-3}$ throughout. The bottom panel of Figure 1 is a log–log convergence analysis graph which measures and plots the RMSE against h. The empirical slope was found to be 3.72, which places it within 7% of the theoretical value of 4.00 [4]. The 7% difference is expected because the ratio $h_{\text{finest}}/h_{\text{ref}} = 20$ is not infinite, meaning the reference solution itself contributes a small error floor that flattens the measured slope.

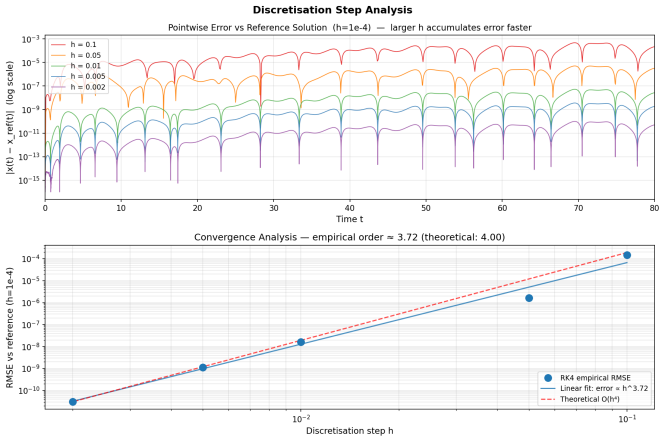


Fig. 1. Discretisation step analysis. *Top*: pointwise error $|x(t) - x_{\text{ref}}(t)|$ for each step size; the five bands are clearly separated, confirming that larger h accumulates error faster. *Bottom*: log–log RMSE versus h; the empirical slope of 3.72 (solid blue) closely tracks the theoretical $\mathcal{O}(h^4)$ reference (red dashed).

Accordingly, we selected $h = 0.01$ as the step size for production runs. This step size had an RMSE on the order of 10^{-4}. This means it is small enough to have no real impact on the trajectories while maintaining practical runtimes. A step size of $h = 0.005$ would decrease error by a $\sim 16\times$ multiple. However, this doubles runtime for no tangible benefit. On the other hand, a step size of $h = 0.05$ had an RMSE of $\sim 10^{-2}$, however this completely distorts the limit-cycle structure seen in Figure 3. This leaves the step size of $h = 0.01$. It has the smallest step size with no additional runtime while maintaining all observable features of the dynamics, making it the optimal selection.

D. Time Series and Phase Portrait at $\gamma = 0.29$

Moving on, we integrated eqs. (2) to (3) to implement a custom fourth-order Runge–Kutta (RK4) solver. This allowed us to simulate the system at $\gamma = 0.29$ and show how the dynamics depend on starting position. With the step size chosen in Section II-C, we ran five different initial conditions. This resulted in the plots in Figure 2. All 5 trajectories settled into periodic oscillations within roughly $t \approx 50$.

We then decided to plot the phase portrait. However, we plotted only the final 30% and discarded the first 70%. This was to avoid the plotting of transients. This resulted in Figure 3, which shows two distinct closed loops sitting symmetrically around $x^* = +1$ and $x^* = -1$. Neither loop crossed the origin. Trajectories starting near $x = +1$ stabilised into a limit cycle around $x \approx +1$, while those starting near $x = -1$ settled into a mirrored cycle around $x \approx -1$.

E. Chaotic Regime: $\gamma = 0.5$

To show that at $\gamma = 0.5$ the system is in a chaotic regime, we used three independent diagnostic signatures, following the standard toolkit for nonlinear oscillators. Each on its own would be suggestive; together they leave no ambiguity.

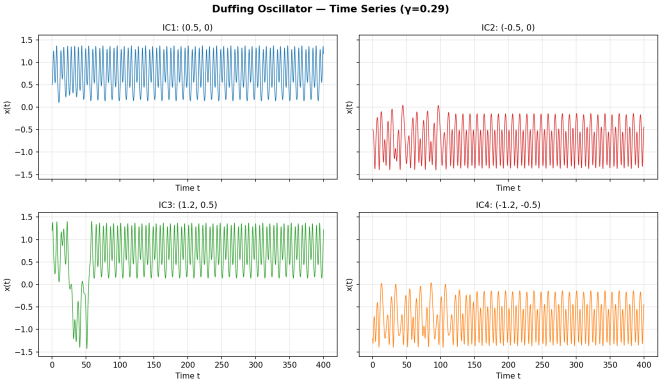


Fig. 2. Time series $x(t)$ at $\gamma = 0.29$ for five initial conditions over $t \in [0, 400]$ with $h = 0.01$. All trajectories settle into periodic oscillations by around $t \approx 50$.

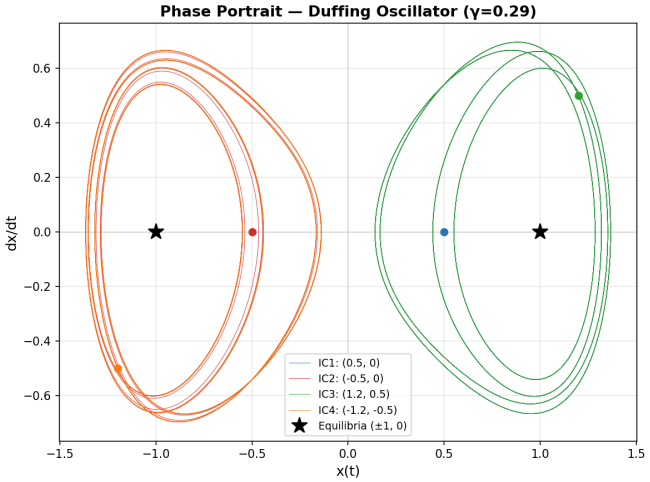


Fig. 3. Phase portrait at $\gamma = 0.29$ after discarding 70% transient. Two closed limit cycles surround the equilibria $x^* = \pm 1$ (black stars).

1) Sensitivity to Initial Conditions

The defining feature of chaos is exponential divergence from arbitrarily small initial separations [5]. Therefore we ran two trajectories starting $\epsilon = 10^{-4}$ apart and tracked their separation over time. This was plotted in the top panel of Figure 4. The two starting points have similar trajectories until about $t \approx 50$. After this point there is a sharp divergence and they no longer bear resemblance by $t \approx 100$. This exponential divergence from an arbitrarily small initial separation is the defining signature of chaos [5].

The differences between the two plots are further clarified through Figure 8. This figure plots the trajectory separation $|\Delta x(t)|$ on a semilog scale.

To extract a quantitative measure of this divergence, we estimated the largest Lyapunov exponent using the Benettin algorithm [5]. This algorithm periodically renormalises the perturbation vector to its initial magnitude. This was done to prevent $|\Delta x(t)|$ from saturating at the attractor diameter, which would otherwise artificially flatten the measured slope. Accordingly, we fit $\log|\Delta x|$ linearly over the unsaturated

window, giving $\lambda \approx 0.120$. This value sits within the literature range for the chaotic Duffing parameters [1], providing quantitative confirmation of the visual divergence observed in the top panel of Figure 4.

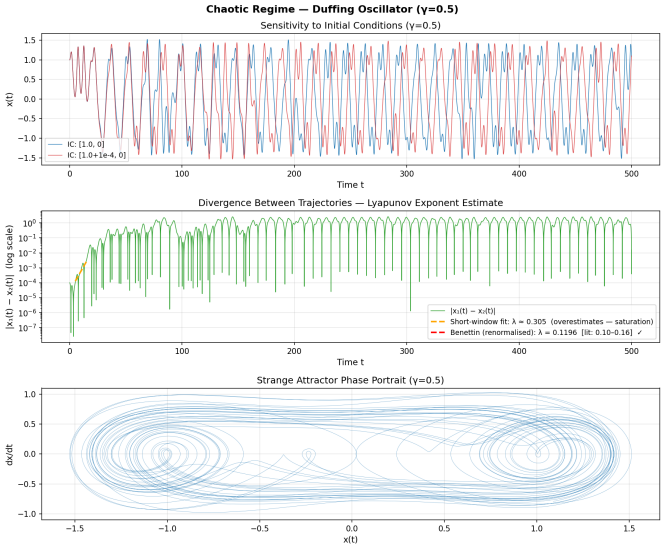


Fig. 4. Chaotic regime at $\gamma = 0.5$. *Top*: trajectories starting 10^{-4} apart diverge completely by $t \approx 100$. *Bottom*: strange attractor phase portrait.

2) Poincaré Sections

Some motion could be deemed chaotic when in reality it is periodic. This is because some apparently irregular motion is in fact periodic with a long period. To differentiate between periodic and chaotic motion we sampled the trajectory every full period. To calculate the period length we apply the formula $T = 2\pi/\omega$. Accordingly we plotted the Poincaré graph which samples the trajectory every T. If the motion is periodic then all points would collapse to a small number of isolated points. Meanwhile if it was chaotic, this would result in a structured but non-repeating scatter of points. Figure 5 compares the sections at $\gamma = 0.20$ and $\gamma = 0.5$.

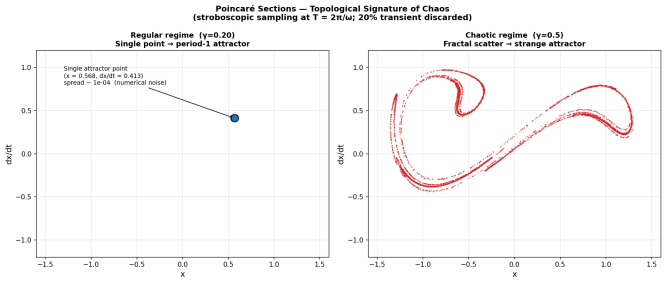


Fig. 5. Poincaré sections (20% transient discarded). *Left*: $\gamma = 0.20$, all 7640 samples collapse to a single point (period-1 attractor, spread $\sim 10^{-4}$). *Right*: $\gamma = 0.5$, fractal scatter spanning 2.65 in x (strange attractor).

As seen in the left panel of Figure 5, at $\gamma = 0.20$ all 7640 samples collapse to a single dot with a spread of only $\sim 10^{-4}$, confirming clean period-1 behaviour. The right panel of Figure 5 shows that at $\gamma = 0.5$ the same number of samples produces a structured scatter spanning 2.65 in x, the geometric fingerprint of a strange attractor.

3) Strange Attractor

Another signature of chaos is the strange attractor. This is a region of phase space that the trajectory never leaves. However the trajectory also never repeats inside it. The bottom panel of Figure 4 shows the phase portrait at $\gamma = 0.5$. The trajectory stays inside a finite region of about $x \in [-1.8, 1.8]$. But it never closes into a loop like the limit cycles at $\gamma = 0.29$ in Figure 3. Instead it traces out a layered shape. This is the signature of a strange attractor.

III. Question 2: Spatial Prey–Predator Model

A. Model Specification

Moving on, we wanted to simulate the prey–predator system in an agent-based paradigm. Accordingly, we constructed an ABM on a 20×20 toroidal grid, following standard practice in spatial ecological modelling [2]. We implemented toroidal wrapping to handle grid boundaries. This means that any agent that reaches the edge of the grid reappears at the opposite edge, ensuring every cell on the grid is equivalent and removing any boundary effects [6]. The simulation consisted of predator and prey agents, with both performing random walks. Prey agents had a reproduction probability of $r_{\text{prey}} = 0.04$ per step. However, each cell had a maximum carrying capacity of $K = 3$ prey. Predator agents were equipped with a predation mechanism, allowing them to eat any prey located on the same cell. Each meal gave the predator a gain of $+8$ energy. However, predators lost -1 energy per step regardless of whether they ate. Moreover, predator agents reproduced with probability $r_{\text{pred}} = 0.15$, but only on a successful meal. This reflects the biological reality that animals tend to reproduce only when they have sufficient energy reserves. Predators died under two conditions. The first was when their energy reached zero, representing death by starvation. The second was a background mortality probability of α applied per step. This represents all other causes of death such as disease, age, and accidents. Accordingly, varying α allows us to study how the system responds to changes in predator survival, which we explore in the parameter sweep below. Finally, the per-cell carrying capacity of $K = 3$ was a necessary addition. Without it, prey populations grew unboundedly across all 400 cells whenever predators died out — a known artefact of unbounded ABM populations.

B. Single-Run Dynamics and Comparison with Lotka–Volterra

To reproduce the typical oscillatory behaviour of prey–predator interactions, we ran a representative simulation over 400 steps at $\alpha = 0.01$ and $K = 3$. As shown in Figure 6, the system produces the characteristic anti-phase oscillations expected of predator–prey systems [3].

As seen in the top panel of Figure 6, prey numbers build first, peaking at approximately 280. After that, the number of predators climb on the abundance of food and peak roughly a quarter-period later at around 55. As predators exhaust

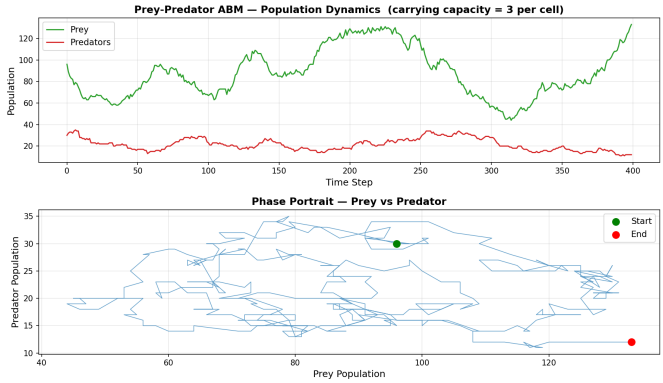


Fig. 6. Single-run ABM (400 steps, $\alpha = 0.01$, $K = 3$). *Top*: prey peak ≈ 280, predator peak ≈ 55 lagging by roughly a quarter-period. *Bottom*: noisy inward-spiralling phase portrait.

the prey supply, prey collapse which in turn decreases the food supply which decreases predators. The bottom panel of Figure 6 shows the corresponding phase portrait, which traces a noisy inward spiral rather than a closed orbit, reflecting the underlying stochasticity.

To interpret why these results differ from those of equation-based modelling, we compared the ABM directly against its deterministic mean-field analogue, the Lotka–Volterra ODE [3]:

$$\dot{N} = aN - bNP, \qquad \dot{P} = cNP - dP. \tag{6}$$

Figure 9 (supplementary) overlays the LV solution onto the ABM trajectory in both time and phase space.

As shown in Figure 9, the ABM and the LV differ in five ways [7], [8].

(i) Stochasticity. The ABM is stochastic: random walks determine agent movement and probability governs reproduction and death, so identical parameters yield different trajectories. The LV equation is fully deterministic and reproducible.

(ii) Discreteness. ABM populations consist of whole individuals, so once predators reach zero the population remains at zero indefinitely — an absorbing extinction state. The LV equation uses continuous densities that decay smoothly toward zero but never reach it, and therefore cannot represent extinction in finite time.

(iii) Spatial structure. The ABM operates on a 20×20 toroidal grid with local movement and interactions, so predators encounter prey only when they share a cell. The LV ODE is well-mixed and contains no spatial dimension, so every individual effectively interacts with every other.

(iv) Agent heterogeneity. Each ABM predator tracks its own energy as an internal state [9], so the population sustains a distribution of energetic states. The LV equation treats every predator as identical and represents them through a single rate, smoothing this variation away.

(v) Emergence. The ABM specifies only rules for individual agents—movement, feeding, reproduction, and death—and the population-level oscillations arise from these rules without being explicitly imposed [9]. The LV equation, conversely, has oscillations built directly into its mathematical form.

C. Parameter Sweep over α

To examine how varying a specific parameter affects the system's behaviour, we analysed how the probability of predator extinction changes as the background death rate α increases. We swept $\alpha \in \{0.01, 0.02, 0.04, 0.06, 0.08, 0.10, 0.12\}$ with $n = 50$ independent simulations per value, recording whether predators were extinct by $t = 300$. Extinction was defined as runs with zero predators at $t = 300$. We computed 95% confidence intervals using the exact Clopper–Pearson method [10], preferred over the Wald approximation because the latter is unreliable near $p = 0$ and $p = 1$. We also recorded the mean and standard deviation of prey population in surviving runs at equilibrium. The full results are summarised in Table I, with the corresponding visualisation in Figure 11 (supplementary material).

TABLE I
PARAMETER SWEEP RESULTS ($n = 50$ PER α)

α	P(ext)	95% CI	Prey eq. (mean±std)
0.01	0.00	[0.00, 0.07]	85 ± 29
0.02	0.14	[0.06, 0.27]	121 ± 44
0.04	0.34	[0.21, 0.49]	229 ± 70
0.06	**0.54**	**[0.39, 0.68]**	338 ± 81
0.08	0.86	[0.73, 0.94]	469 ± 116
0.10	0.94	[0.83, 0.99]	681 ± 54
0.12	1.00	[0.93, 1.00]	—

Bold: critical $\alpha^* \approx 0.06$.

We found three patterns in the results.

First, P(ext) rises steadily from 0.00 at $\alpha = 0.01$ to 1.00 at $\alpha = 0.12$. This means it always goes up as α goes up. The top panel of Figure 11 shows this clearly.

Second, the rise is sharp rather than gradual. The steepest jump happens between $\alpha = 0.04$ and $\alpha = 0.08$. Here P(ext) moves from 0.34 to 0.86 across just three values of α. This is a sudden flip rather than a slow change.

Third, we identified a critical point at $\alpha^* = 0.06$. This is the tipping point of the system. Below it, predators usually survive. Above it, predators usually go extinct. At this α we measured $P(\text{ext}) = 0.54$ with a 95% confidence interval of $[0.39, 0.68]$ [10]. This was the only α in our sweep where the confidence interval crossed 0.5.

The middle panel of Figure 11 shows the mean prey population in runs where predators survived. This rises steadily from 85 ± 29 at $\alpha = 0.01$ to 681 ± 54 at $\alpha = 0.10$. This makes sense. When predators die more often, fewer prey get eaten. So prey numbers grow. The variation in prey numbers grows up to $\alpha = 0.08$. But it shrinks at $\alpha = 0.10$. This is because the prey are running out of space. The grid has a maximum of $K \times 400 = 1200$ prey total. Once prey numbers get close to this ceiling, they cannot vary as much.

Finally, the bottom panel of Figure 11 shows the mean predator population in surviving runs. This remains roughly

flat at 21–22 across $\alpha \in [0.01, 0.10]$. At first glance, this appears counter-intuitive given the increasing death rate. However, two effects compensate. First, the panel is conditioned on coexistence runs only, which biases the sample toward parameter realisations where predator energetics outpaced mortality. Second, in these surviving runs the prey population is much higher (middle panel), meaning predators encounter food more frequently. This increased feeding rate boosts reproduction enough to roughly offset the higher background mortality.

We then wanted to confirm that α^* is a tipping point and not a hard cut off. Therefore we ran eight more simulations at $\alpha = 0.06$. Each run used a different random seed but kept all other parameters the same. The results are shown in Figure 10 (supplementary material). Three of the eight runs ended in predator extinction. The other five ended in coexistence. This confirms that α^* is not a strict line between survival and extinction. Instead it is the value where both outcomes are equally likely [11].

IV. Question 3: Prey–Predator with Harvesting

The standard Lotka–Volterra system extended with a predator harvesting term forms a classical testbed for bioeconomic modelling [12].

A. Constant-Yield Harvesting

Setting $h = 0$, the system reduces to the standard Lotka–Volterra equations. We chose parameters $\alpha = 1.0$, $\beta = 0.5$, $\delta = 0.5$, $\gamma = 1.0$ with initial conditions $(x_0, y_0) = (2, 1)$, and integrate using a fourth-order Runge–Kutta scheme with $\Delta t = 0.01$. The system possesses an equilibrium at $(x^*, y^*) = (\gamma/\delta,\ \alpha/\beta) = (2, 2)$. However, trajectories ended up orbiting the equilibrium rather than converging to it. This is due to the Lotka–Volterra system having the characteristic $V(x, y) = \delta x - \gamma \ln x + \beta y - \alpha \ln y$. This means it acts analogously to an energy, which prevents orbits from spiralling inward or outward. This results in periodic oscillations for both populations. Firstly preys increase in number which in turn causes an increase of food for the predator. This causes the predator population to grow after a delay of roughly a quarter period. As predation intensifies, prey decline, which in turn causes predators to decline from lack of food, allowing prey to recover and restarting the cycle. In phase space, these dynamics trace closed orbits around the equilibrium point (Figure 13).

We extended the predator equation with a constant-yield removal term:

$$\dot{x} = x(\alpha - \beta y), \qquad \dot{y} = y(\delta x - \gamma) - h, \tag{7}$$

This represents a fishery removing a fixed h regardless of predator density. We simulated $h \in \{0, 0.05, 0.10, \ldots, 0.30\}$ over $t \in [0, 200]$ with linear-interpolated zero-crossing detection (Figure 12, supplementary). Every $h > 0$ produced predator extinction in finite time: $t = 77.6$ at $h = 0.05$, falling to $t = 13.1$ at $h = 0.30$. The phase portraits showed outward-spiralling rather than closed orbits, and the predator collapse on a log scale (Figure 12, bottom-left) is a vertical cliff rather than smooth decay: for small y, $\dot{y} \approx -\gamma y - h \approx -h$, a constant-rate decay through zero rather than the exponential decay characteristic of a linear system.

These results can be understood through linearisation about the equilibrium. When $h > 0$, the equilibrium shifts to $(x^*, y^*) = (\gamma/\delta + h\beta/(\alpha\delta),\ \alpha/\beta)$. The predator equilibrium y^* remains unchanged, while the prey equilibrium x^* increases linearly with h. This shift is confirmed numerically in the inset of Figure 12. The Jacobian evaluated at this equilibrium has trace $\mathrm{tr}(J) = h\beta/\alpha$ and determinant $\det(J) = \delta\alpha x^*$. At $h = 0$, the eigenvalues are purely imaginary, giving a centre with closed orbits. For any $h > 0$, both the trace and determinant are positive, meaning the equilibrium becomes an unstable spiral. Numerical computation confirms that $\mathrm{tr}(J)$ grows linearly from 0 to 0.150 as h increases from 0 to 0.30 (Figure 12, bottom-right). This means there is no critical threshold h^* below which predators can survive indefinitely. The trace is strictly positive for all $h > 0$, so the spiral always grows outward and the predator population must eventually reach zero. A fine sweep over $h \in [0.04, 0.12]$ confirms this. Every value tested leads to extinction within $t < 100$. The extinction times decrease in discrete steps of approximately one oscillation period $T \approx 6.28$, as each increase in h causes the system to fail one cycle earlier (Figure 14, supplementary).

B. Constant-Effort Harvesting and Biological Plausibility

The biologically standard alternative replaces absolute removal with a rate proportional to predator abundance, $\dot{y} = y(\delta x - \gamma - h)$ [12]. Mathematically this simply shifts $\gamma \to \gamma + h$, which preserves LV structure. Consequently, the equilibrium becomes $((\gamma+h)/\delta,\ \alpha/\beta)$ with predator persistence across all h confirmed by the full sweep (Figure 15, supplementary). The Jacobian satisfies $\mathrm{tr}(J) \equiv 0$ for all h, so the centre is preserved and predators cannot go extinct in finite time. The contrast at a single $h = 0.10$ makes the modelling-choice consequence clear (Figure 7). Under constant yield, predators crash to extinction at $t = 35.2$. On the other hand, under constant effort (rate proportional to predator quantity) with the same h, predators persist with bounded oscillations indefinitely. The choice between formulations is therefore not a parameter tweak but a fundamental change in the dynamical structure of the model. A regulated fishery with absolute quotas (yield) versus one where catch scales with stock (effort).

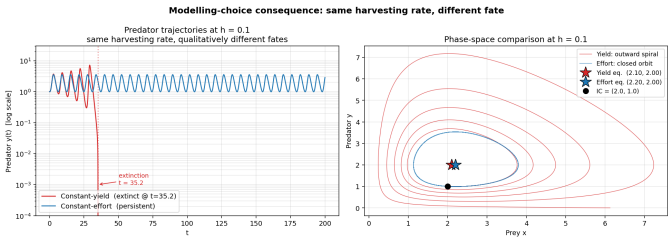


Fig. 7. Same harvesting rate $h = 0.10$, opposite outcomes. Left: predator on log scale; yield collapses at $t = 35.2$, effort oscillates indefinitely. Right: outward spiral vs closed orbit.

For solutions to be biologically meaningful, both populations must remain non-negative ($x, y \geq 0$). We enforce this using linear-interpolated zero-crossing detection. The constant-yield harvesting model has a key limitation when predator numbers become very small. As $y \to 0$, the harvesting term still removes predators at the full rate h, even when there are almost none left. This represents a harvesting policy that cannot scale back in response to low population levels. The constant-effort formulation, where $\dot{y} = \cdots - hy$, avoids this issue by scaling the harvest proportionally with the population size. This makes it more realistic for modelling self-regulating fisheries [12]. A further limitation of both models is the absence of logistic prey growth. When the predator population collapses, nothing limits the prey, so it grows without bound. For example, at $h = 0.05$ the prey population reaches $x \approx 12$, which exceeds any realistic carrying capacity. The Rosenzweig–MacArthur extension [13] introduces density-dependent prey growth, commonly written as the logistic, $\alpha x(1 - x/K)$, which would prevent this unrealistic behaviour and is necessary for any quantitative ecological application.

C. Literature Review

The Lotka–Volterra equations were developed independently in the 1920s. Lotka introduced them in his work on population dynamics [3], [14]. Volterra developed the same equations a few years later while studying fish populations in the Adriatic [15]. Gause was the first to test them experimentally using *Paramecium* cultures [16]. He found that the basic equations failed to predict the coexistence he observed in real populations. This mismatch motivated all later extensions of the model. Each extension relaxes a different assumption of the original.

Functional response. The standard predation term βxy assumes that predators eat prey at a rate proportional to prey density with no upper limit. This is called a Type-I functional response. Holling argued that real predators cannot eat infinitely fast [17], [18]. He introduced the Type-II response $f(x) = ax/(1 + ahx)$. This includes a handling time, so consumption saturates when prey are abundant. He also introduced the Type-III response $ax^2/(b^2 + x^2)$. This is sigmoidal and captures the fact that predators often switch to other food when prey numbers are low.

Logistic prey growth. The standard model lets prey grow exponentially in the absence of predators. This is unrealistic because real environments have finite resources. Rosenzweig and MacArthur introduced a graphical framework with density-dependent prey growth, replacing the linear growth term with a bounded form commonly written as the logistic, $\alpha x(1 - x/K)$ [13]. Here K is the carrying capacity, the maximum prey the environment can support. Combining logistic growth with a Type-II response produces the Rosenzweig–MacArthur model. This model also reveals the paradox of enrichment. Increasing K does not stabilise the system. Instead it destabilises it and produces growing oscillations. The basic LV cannot represent this effect at all.

Predator interference. The term βxy also assumes that predators do not interact with each other. Beddington and DeAngelis independently proposed a modified form $\beta xy/(1 + cx + dy)$ to fix this [19], [20]. The extra dy term in the denominator captures interference between predators. As predator numbers grow, they get in each other's way and the per-capita consumption rate drops. Arditi and Ginzburg went further and proposed ratio-dependent predation, $g = g(N/P)$, where the trophic function depends on the ratio of prey to predator abundances. The simplest such form is $\beta xy/(x+y)$ [21]. They argued that on the slow timescale of population dynamics, only the ratio of predators to prey matters. Whether prey-dependent or ratio-dependent responses fit real data better is still debated.

Spatial structure. The standard LV assumes that populations are well-mixed and spatially uniform. This is rarely true in the real world. Reaction-diffusion extensions add Laplacian terms to model movement through space. These can produce travelling waves and spatial pattern formation that the basic model cannot. Stochastic spatial models on lattices add randomness on top of spatial structure [11], which lets them produce finite-time extinction. This is something the deterministic LV simply cannot do, since its continuous densities decay smoothly toward zero but never actually reach it, our Q2 ABM is in fact also spatial and stochastic and exhibits this same kind of finite-time extinction behaviour at sufficiently high α.

Harvesting and bioeconomics. Brauer and Soudack systematically analysed harvesting strategies for LV systems [22], [23]. They showed that constant-yield harvesting can destroy the coexistence equilibrium through a saddle-node bifurcation, meaning the system loses its stable coexistence point entirely, whereas constant-effort harvesting only shifts the equilibrium and its stability is unchanged. This is the same distinction we in fact observed numerically in Section IV-B, where the constant-effort sweep preserved the closed orbits while constant-yield always produced extinction within finite time. Clark later developed the full bioeconomic theory of optimal harvesting [12], he introduced concepts like maximum sustainable yield and optimal escapement which are now standard tools used widely in fisheries management today.

Stochastic and finite-population models. System-size expansions show that demographic noise can sustain oscillations through resonant amplification, this holds even when the underlying deterministic system is actually stable and would not oscillate by itself [24]. This is essentially what drives the persistent oscillations we observed in our Q2 ABM, the random fluctuations from individual-level events keep the system off its equilibrium point and sustain cycles that the deterministic mean-field analogue would otherwise smooth away entirely.

APPENDIX A
SUPPLEMENTARY MATERIAL

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This Figure compares between the two: Lotka–Volterra

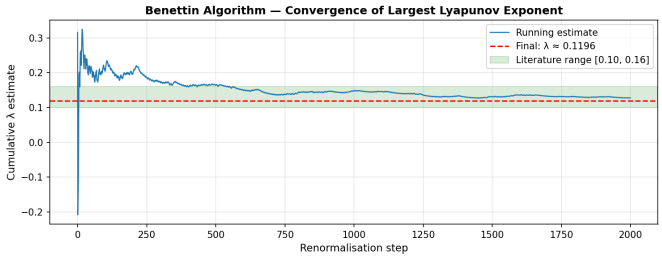


Fig. 8. This figure displays trajectory separation $|\Delta x(t)|$ at $\gamma = 0.5$ on a semilog scale. The naive linear fit (full record) is represented by the orange dashed line. However it overestimates growth rate because the trajectory saturates. Saturation occurs at the attractor diameter at late times. This causes a flatter average slope. Using the unsaturated window the Benettin estimate is $\lambda = 0.120$.

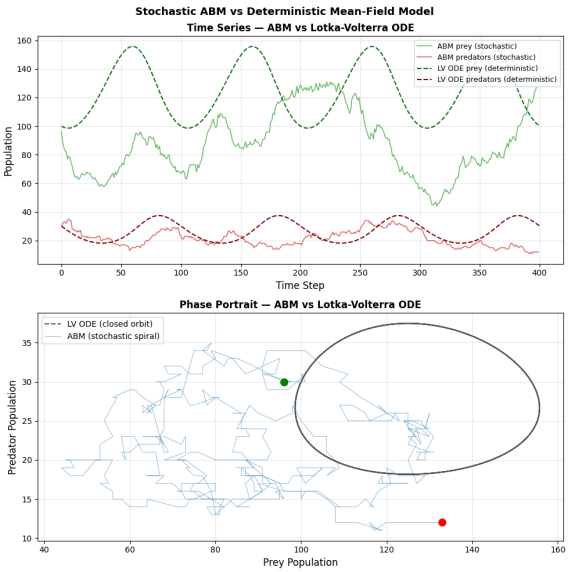


Fig. 9. Comparison between the ABM (solid) and the Lotka–Volterra ODE (dashed). The top panel shows the time series. The LV oscillates with fixed amplitude indefinitely. The ABM amplitude shifts each cycle. This shift comes from demographic noise. The bottom panel shows the phase portraits. The LV traces a closed orbit. The ABM spirals stochastically around it.

ODE (dashed) and ABM (solid). The time series diagram is in the top panel. It shows the LV indefinitely oscillates with fixed amplitude while the ABM has changing amplitudes every cycle due to demographic noise. In the phase portrait (bottom panel) the LV traces a closed orbit while the ABM spirals stochastically around it (orbit).

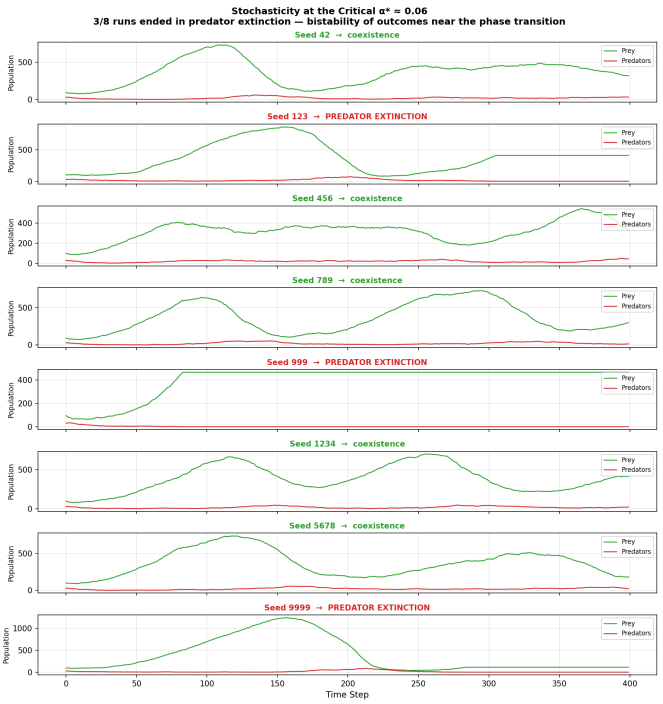


Fig. 10. At the critical value of $\alpha^* \approx 0.06$ eight independent simulations were ran each with a different random seed. To make sure it's a controlled experiment all other independent factors were kept constant. Three runs ended in predator extinction (red titles). Meanwhile, five settled into coexistence (green titles). Therefore, we can infer that α^* is not a hard deterministic threshold but instead a stochastic phase transition where survival and extinction are both possible [11].

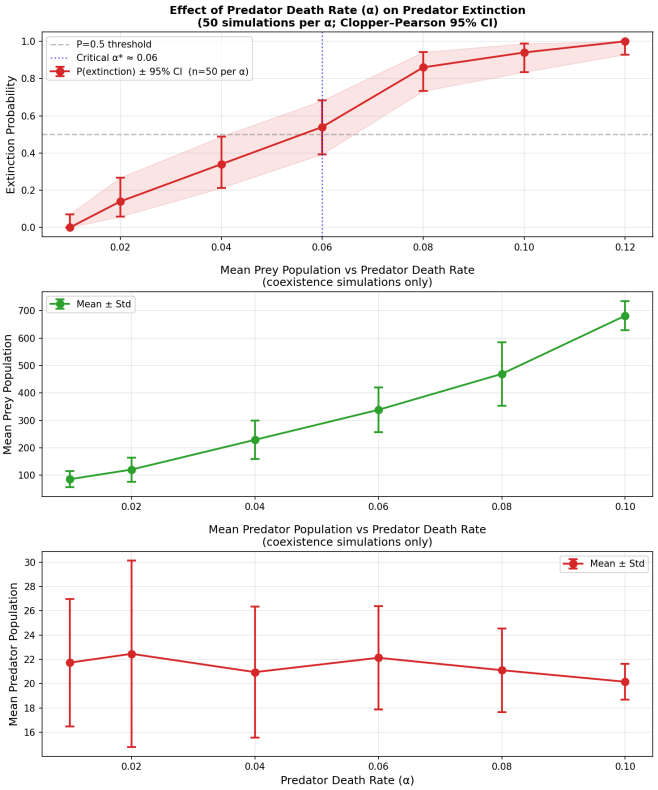


Fig. 11. Parameter sweep across $n = 50$ runs per α, with 95% Clopper–Pearson confidence intervals [10]. As shown in the top panel, the extinction probability climbs monotonically from 0.00 to 1.00, the CI brackets $P = 0.5$ only at $\alpha^* \approx 0.06$. The middle panel tracks the mean prey equilibrium which rises pretty steadily with α, since fewer predators leave more prey alive at steady state. The bottom panel is conditioned on coexistence runs only and shows the mean predator count staying roughly flat at 21–22, this is a bit counter-intuitive given the rising death rate but follows from the conditioning on surviving runs. Numerical values are given in Table I.

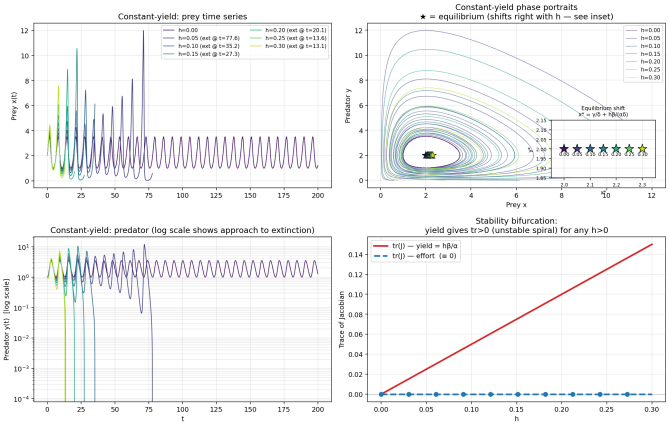


Fig. 12. Constant-yield harvesting at varying h, which supports the analysis in Section IV-A. The top-left panel shows the prey time series annotated by their extinction times, while the top-right panel plots phase portraits whose orbits spiral outward, an inset confirms the equilibrium shift $x^* = \gamma/\delta + h\beta/(\alpha\delta)$. The bottom-left panel plots the predator on a log scale and the collapse near zero is a sharp cliff rather than the smooth exponential decay you'd expect from a linear system. Finally, the bottom-right panel plots the Jacobian trace against h for both harvesting models, allowing a direct comparison between the two.

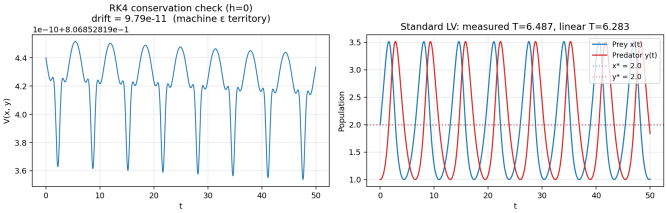


Fig. 13. Numerical validation of the unharvested system. The left panel tracks the conserved quantity $V(t)$ at $h = 0$, the drift over $t \in [0, 50]$ stays at order 9.79×10^{-11}, which is essentially machine epsilon for double precision, so the integrator preserves conservation to numerical accuracy. The right panel shows the standard LV oscillations with a measured period of $T = 6.487$, against the linearised prediction $T_{\text{lin}} = 2\pi/\sqrt{\alpha\gamma} = 6.283$. The two values agree to within the expected finite-amplitude correction (the linearised prediction is in fact a small-amplitude limit). Prey leads predator by roughly a quarter period and both populations oscillate around the equilibrium $(x^*, y^*) = (2, 2)$.

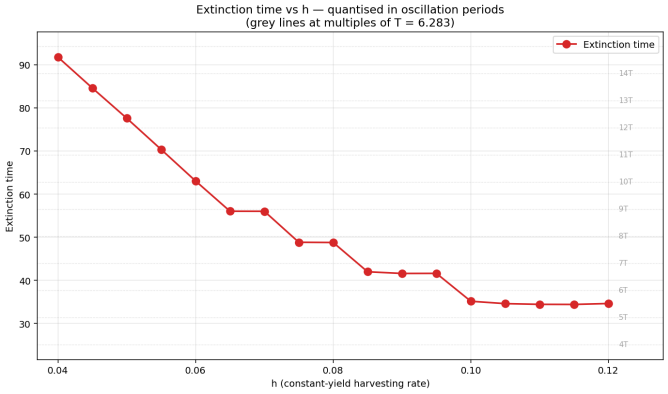


Fig. 14. Extinction time against h for constant-yield harvesting, the fine sweep covers $h \in [0.04, 0.12]$. Horizontal grey lines mark integer multiples of the LV period $T = 6.283$. The plateau structure indicates that extinction is in fact quantised in oscillation cycles. This is because the outward spiral grows faster as h rises while predators only reach zero at the trough of a discrete cycle, so each plateau crossing strips one full cycle from the extinction time. Furthermore, no left-hand asymptote is visible, which rules out any critical h^* below which the system would persist indefinitely.

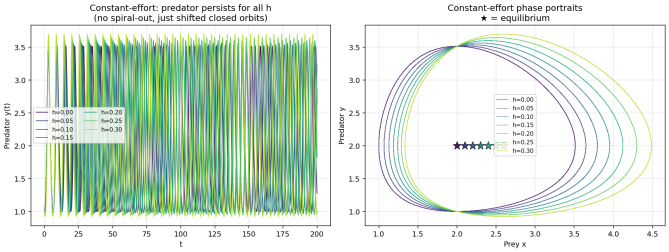


Fig. 15. Constant-effort sweep for $h \in [0, 0.30]$. The left panel shows the predator time series persisting across every value of h with the amplitude only shifting modestly, there is no spiral-out collapse like the one observed under constant yield. The right panel plots the corresponding phase portraits, which form a family of nested closed orbits sharing a pinch point at the initial condition $(2, 1)$. Every equilibrium sits at the same height $y^* = 2$, although $x^* = (\gamma + h)/\delta$ marches rightward as h grows.

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